

I. Statement of Work

1.0 Purpose: To support the research and development of the Passive fuel-Cell Surface-power System (PaCeSS) effort awarded via the JSC EA Engineering Innovation Fund (EIF) for FY'25.

2.0 Scope: This procurement addresses the research and development commitment from Texas A&M University associated with the PaCeSS award. Texas A&M Engineering Experiment Station (TEES) will model, predict, and report on behavior of the two-phase thermosyphon driven actuating radiator that is being developed for the PaCeSS EIF project. An actuator will be provided for the purpose of developing a mechanical and fluid interface for integration with the PaCeSS test article built at JSC. The same actuator and support will be provided for an integrated benchtop test to be performed at JSC no later than September 2025. This effort leverages TAMU's previous work with Shape Memory Alloys (SMA) and actuating radiators.

3.0 References:

Texas A&M Engineering Experiment Station (TEES) Letter of commitment dated February 5, 2025
PaCeSS proposal

4.0 Requirements:

4.1 Performance:

- 4.1.1 Shall demonstrate through analysis and testing the operation of a prototype actuating radiator within which a two-phase water thermosyphon carries thermal energy and which is fully closed at a temperature no less than 50C and fully open at a temperature no greater than 90C.

5.0 Administrative:

5.1

6.0 Contractor Tasks and Deliverables:

The Contractor shall perform the following tasks:

- 6.1 Task 1: Predict the behavior of a two-phase thermosyphon driven actuating radiator as integrated into the PaCeSS concept.
- 6.2 Task 2: Report on assumptions, parameters, and result of this prediction at the conclusion of the period of performance.
- 6.3 Task 3: Provide to JSC an actuator for the purpose of developing and demonstrating a mechanical and fluid interface to the PaCeSS concept hardware. This actuator will be returned following integrated testing.
- 6.4 Task 4: Support an integrated benchtop test to be performed at JSC no later than September 2025

The Contractor shall provide the following deliverables:

- 6.5 A torque tube actuator for sizing and test integration. Actuator will be returned following integrated testing.
- 6.6 A final report detailing analysis, development, and results of work performed.

7.0 Period of Performance:

Required items shall be delivered according to the following schedule unless reviewed and approved by contract modification.

- 7.1 Midterm Milestone: Torque tube actuator – NLT May 15, 2025
- 7.2 Final Milestone: Final Report – NLT September 30, 2025